

Reference-Assisted Phase Retrieval for Holographic MIMO Reception: Unique Channel and Finite-Alphabet Symbol Recovery

Abstract—This paper develops a reference-assisted phase-retrieval framework for holographic MIMO reception, jointly addressing continuous channel sensing and finite-alphabet data recovery from intensity-only measurements. A controllable RF reference is superimposed element-wise with the incident field before square-law detection, breaking the global-phase invariance without coherent I/Q acquisition. For the resulting affine intensity model, we establish an exact secant identity and derive Jacobian-based identifiability and stability conditions for continuous channel parameters. For finite-alphabet signaling, we show that noiseless symbol recovery is unique if and only if the minimum energy-domain distance is positive, and derive corresponding pairwise-error and stability bounds. A spatial four-phase reference code yields an average separation guarantee and reveals that severe reference-signal energy imbalance weakens phase information. Guided by these results, a two-stage receiver combines linearized intensity-domain initialization, stability-guided adaptive training, damped Gauss-Newton channel estimation, and distance-certified Gerchberg-Saxton symbol detection. Simulations verify the theoretical conditions and show that, at 10 dB, the adaptive policy uses 20.0 pilot symbols on average instead of the fixed budget of 40 while attaining a channel normalized mean-square error (NMSE) of -28.1 dB; the proposed distance metric also predicts finite-alphabet detection reliability.

Index Terms—Holographic MIMO, reference-assisted phase retrieval, channel identifiability, finite-alphabet symbol recovery, intensity-only measurements.

I. INTRODUCTION

Reconfigurable intelligent surfaces (RISs) provide a programmable electromagnetic interface for controlling wireless propagation with low-complexity surface elements. Their operating principles, communication models, and potential gains in coverage and energy efficiency have been extensively studied [1]–[9]. Beyond conventional reflecting architectures, densely sampled large intelligent and holographic surfaces can act as spatially continuous apertures, offering high spatial resolution and new wave-domain processing capabilities [10]–[13]. Recent studies have further explored holography-inspired surface control, wave-domain DOA estimation, and metasurface-enabled sensing and localization [14]–[18]; wavenumber-domain processing provides a complementary framework for modeling and processing quasi-continuous holographic apertures [19].

A central implementation challenge is that acquiring the complex field at every surface element conventionally requires coherent radio-frequency chains, local-oscillator distribution, and in-phase/quadrature sampling. Power-detector-based architectures instead retain only intensity measurements and can substantially simplify the receiving hardware [20], [21]. Square-law or direct detection is well established in optical

and infrared communication systems [22], [23]; when used for complex-valued radio signals, however, it discards the phase that carries essential channel and symbol information. The resulting inverse problem is therefore a form of phase retrieval.

Phase retrieval has a long history in optics and signal processing [24], [25]. Convex lifting and semidefinite relaxations provide recovery guarantees at the cost of high-dimensional optimization [26]–[28], whereas alternating-minimization and gradient-based methods offer more scalable nonconvex alternatives [29], [30]. Recent deep-unfolding approaches further combine interpretable iterative structure with learned signal priors [31]. In parallel, fundamental results have characterized the number and structure of measurements required for uniqueness and stable recovery [32]–[36]. These results clarify the generic phaseless inverse problem, but its unavoidable global-phase ambiguity remains incompatible with the direct recovery of absolute communication symbols.

A known reference wave can remove this ambiguity by interfering with the unknown field before square-law detection. This principle originates from holography, where the object and reference waves form an intensity pattern that encodes their relative phase [37], [38]; controlled reference phases can provide additional complex-field information [39]. Algebraically, the reference converts homogeneous magnitude measurements into affine magnitude measurements, connecting the receiver considered here to affine phase retrieval [40]. Unlike a conventional coherent local oscillator used after antenna combining, the reference in this architecture participates directly in the element-wise interference pattern.

Phaseless recovery has also been investigated in communication problems such as OFDM channel estimation [41]. For holographic interference surfaces, prior work has exploited the spatial sum-of-exponentials structure to enable phase-shifting interference suppression and Prony-based multi-user channel segmentation [21]. A subsequent wideband extension further introduced holographic channel inversion together with maximum-likelihood and low-complexity sensing algorithms [42]. As is typical for Prony-type estimators, the robustness of such methods under a finite aperture is closely related to the conditioning of the associated pole-recovery and Vandermonde systems, particularly when the spatial frequencies of different users are closely spaced. Meanwhile, RIS channel acquisition and codebook design have been extensively studied under conventional complex-baseband observation models [43]–[45]. These studies provide effective channel-sensing and training mechanisms, but they do not characterize the finite-measurement injectivity and stability of the complete affine-intensity mapping, nor do they address the identifiability of

finite-alphabet symbols. In particular, generic recovery guarantees for continuous-valued signals do not directly reveal when two distinct constellation vectors may produce identical or nearly indistinguishable energy patterns under the joint effects of the reference wave, waveform sampling, and the holographic array manifold.

This paper addresses that gap through a unified reference-assisted phase-retrieval framework for RF holographic reception. We first separate field-domain injectivity from reference-induced intensity separation and derive conditions under which finite-alphabet symbols are uniquely represented by the observed energies. We then introduce an energy-domain minimum-distance metric that quantifies noise robustness and exposes the joint roles of reference coding and spatial sampling. Finally, we develop practical channel-estimation and symbol-detection algorithms whose performance can be interpreted through these identifiability and separation properties.

The main contributions of this paper are summarized as follows:

- We formulate a reference-assisted RF holographic receiver and a unified affine-intensity model for the training and data phases, in which a spatially coded reference enables complex channel and symbol recovery without coherent I/Q acquisition.
- For continuous channel sensing, we establish an exact secant identity and Jacobian-based uniqueness and stability bounds, and derive parametric rank and measurement-count requirements that expose the roles of pilot, aperture, and reference diversity.
- For finite-alphabet detection, we characterize noiseless uniqueness through the energy-domain minimum distance, derive pairwise and block-error bounds, and quantify how spatial four-phase coding and reference-signal energy allocation affect separation.
- Guided by these results, we develop a two-stage receiver combining linearized initialization, stability-guided adaptive training, damped Gauss-Newton channel estimation, and distance-certified reference-assisted Gerchberg-Saxton detection.

The remainder of this paper is organized as follows. Section II presents the reference-assisted RF holographic receiver and its unified training and data models. Section III establishes the uniqueness, stability, and finite-alphabet reliability results. Section IV develops the corresponding channel-estimation and symbol-detection algorithms and derives the DOA CRLB. Section V reports the numerical results, and Section VI concludes the paper.

II. SYSTEM MODEL

We consider an RF holographic MIMO receiver in which L single-antenna users transmit complex baseband signals to a surface composed of multiple receiving units. At each unit, the incident RF field is coherently combined with a known reference derived from a common local oscillator before square-law detection, so that phase information is embedded into the resulting intensity measurements [21], [46]. Unlike a conventional passive RIS, the surface operates as an active

receiving front end rather than as a reconfigurable reflector. The receiver first estimates the channel parameters and directions of arrival (DOAs), and then detects the transmitted data symbols using the estimated channel.

The scheduled users are assumed to be frame- and symbol-synchronous at the receiver through standard uplink synchronization and timing-control procedures [47], [48]. Residual timing and carrier-frequency offsets are assumed sufficiently small to have negligible impact over one detection interval; asynchronous users with unknown relative delays are outside the scope of this model.

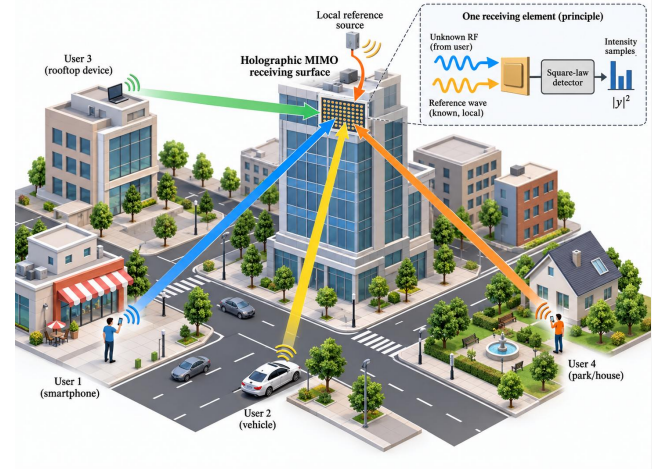


Fig. 1. Illustration of the proposed multi-user reference-assisted RF holographic reception architecture with intensity-only measurements.

A. Transmitter Model

The K symbols transmitted by user l in one signaling block are collected in $\mathbf{s}_l = [s_{l,0}, \dots, s_{l,K-1}]^T \in \mathcal{S}^K$, where $\mathcal{S} = \{c_0, \dots, c_{M_{\text{mod}}-1}\} \subset \mathbb{C}$ is an equiprobable constellation normalized to $\frac{1}{M_{\text{mod}}} \sum_{q=0}^{M_{\text{mod}}-1} |c_q|^2 = \mathbb{E}\{|s_{l,k}|^2\} = 1$. After pulse shaping, the complex baseband waveform is

$$x_l(t) = \sqrt{P_l} \sum_{k=0}^{K-1} s_{l,k} p(t - kT_s), \quad (1)$$

Here, P_l is the transmit power, $p(t)$ is the pulse-shaping filter, and T_s is the symbol duration. The corresponding real passband waveform radiated by user l is

$$x_l^{\text{RF}}(t) = \sqrt{2} \text{Re}\{x_l(t) e^{j2\pi f_c t}\}, \quad (2)$$

where f_c is the carrier frequency. Under the usual carrier-period averaging, the factor $\sqrt{2}$ makes the average power of $x_l^{\text{RF}}(t)$ equal to the envelope power $|x_l(t)|^2$. Thus, $x_l(t)$ contains the pulse-shaped amplitude and phase information, while the carrier term translates that waveform to RF for propagation.

B. Far-Field Channel Model

The RF holographic receiving surface is modeled as an $N_y \times N_z$ uniform planar array (UPA) with a total of $N_r = N_y N_z$ receiving elements. Without loss of generality, the

array is assumed to lie on the yz -plane, with its geometric center located at the origin. The inter-element spacings along the y - and z -axes are denoted by d_y and d_z , respectively. Accordingly, the position vector of the (n, m) -th receiving element is

$$\mathbf{r}_{n,m} = [0, y_n, z_m]^T, \quad (3)$$

where $y_n = \left(n - \frac{N_y+1}{2}\right) d_y$, $n = 1, \dots, N_y$, and $z_m = \left(m - \frac{N_z+1}{2}\right) d_z$, $m = 1, \dots, N_z$. This centered coordinate convention is adopted throughout the paper for the subsequent far-field array-response modeling. The centered coordinates introduce no loss of generality because a change of origin produces only a user-dependent phase that can be absorbed into the complex gain.

Under far-field propagation, user l produces a plane wave with polar angle θ_l measured from the array broadside and azimuth angle ϕ_l measured from the positive y -axis in the yz -plane. With $k_0 = 2\pi/\lambda$, the relevant tangential wavenumbers are $k_{l,y} = k_0 \sin \theta_l \cos \phi_l$ and $k_{l,z} = k_0 \sin \theta_l \sin \phi_l$. The phase at element (n, m) is

$$\psi_{n,m}^{(l)} = \mathbf{k}_l^T \mathbf{r}_{n,m} = k_{l,y} y_n + k_{l,z} z_m. \quad (4)$$

The far-field amplitude is taken as constant across the aperture, while path loss, shadowing, propagation phase, and the residual constant carrier-phase offset relative to the receiver reference are absorbed into $\alpha_l \in \mathbb{C}$. Accordingly,

$$\mathbf{h}_l = \alpha_l \mathbf{v}_l, \quad [\mathbf{v}_l]_{\iota(n,m)} = e^{j\psi_{n,m}^{(l)}}, \quad (5)$$

where $\iota(n, m) = (n-1)N_z + m$. Collecting all users gives

$$\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_L] = \underbrace{[\mathbf{v}_1, \dots, \mathbf{v}_L]}_{\mathbf{V}} \underbrace{\text{diag}(\alpha_1, \dots, \alpha_L)}_{\mathbf{\Gamma}} = \mathbf{V}\mathbf{\Gamma}, \quad (6)$$

which separates the directional array response from the complex path gains. The single dominant far-field path per scheduled user is adopted to expose the DOA-dependent parametric structure targeted by the estimator. More general wideband or multipath fields remain compatible with the affine intensity observation model, but require an expanded channel parametrization and are not considered here.

C. Reference-Assisted Holographic Reception Model

The element-wise reference signals are derived from a common local oscillator operating at f_c and distributed across the surface through a power-divider, RF feed lines, and prescribed phase-delay sections [21], [46]. Thus, all reference branches share the same carrier and maintain stable relative phases over each detection interval. Fixed amplitude and phase offsets introduced by the distribution network are assumed to be calibrated and included in the prescribed reference coefficients. Let $\mathbf{x}(t) = [x_1(t), \dots, x_L(t)]^T$. At receiving unit i , the resulting reference is

$$b_i^{\text{RF}}(t) = \sqrt{2} \text{Re}\{b_i e^{j2\pi f_c t}\}, \quad b_i = \rho_i e^{j\chi_i}, \quad (7)$$

where ρ_i and χ_i are its prescribed amplitude and phase. Let $\mathbf{b} = [b_1, \dots, b_{N_r}]^T$ collect the reference coefficients. In the spatial four-phase implementation, distinct receiving units are

assigned $\chi_i \in \{0, \pi/2, \pi, 3\pi/2\}$ and operate simultaneously. Hence, the four phases form a calibrated spatial code across the aperture; they are not four sequential observations of the same field coordinate and incur no additional symbol intervals.

Under the narrowband model, the received user field has envelope $[\mathbf{H}\mathbf{x}(t)]_i$. Its superposition with (7) gives

$$r_i^{\text{RF}}(t) = \sqrt{2} \text{Re}\{([\mathbf{H}\mathbf{x}(t)]_i + b_i) e^{j2\pi f_c t}\}. \quad (8)$$

Here, $\mathbf{H}\mathbf{x}(t)$ contains the propagation response and unknown user phases, whereas b_i is receiver controlled. The reference-assisted interference is therefore formed in the analog RF domain before square-law detection and sampling.

Each symbol may be observed over M temporal subslots of duration $\Delta_t = T_s/M$, with centers $t_{k,m} = kT_s + (m + 1/2)\Delta_t$ and intervals $\mathcal{I}_{k,m} = [kT_s + m\Delta_t, kT_s + (m+1)\Delta_t]$. Direct square-law detection and integration are applied to the RF waveform in (8). For compactness, denote its composite complex envelope by $u_i(t) \triangleq [\mathbf{H}\mathbf{x}(t)]_i + b_i$. Then

$$\begin{aligned} z_i[k, m] &= \frac{1}{\Delta_t} \int_{\mathcal{I}_{k,m}} [r_i^{\text{RF}}(\tau)]^2 d\tau + w_i[k, m] \\ &= \frac{1}{\Delta_t} \int_{\mathcal{I}_{k,m}} |u_i(\tau)|^2 d\tau \\ &\quad + \frac{1}{\Delta_t} \text{Re}\left\{ \int_{\mathcal{I}_{k,m}} u_i^2(\tau) e^{j4\pi f_c \tau} d\tau \right\} + w_i[k, m] \\ &\simeq \frac{1}{\Delta_t} \int_{\mathcal{I}_{k,m}} |u_i(\tau)|^2 d\tau + w_i[k, m]. \end{aligned} \quad (9)$$

After mean-noise-floor calibration, thermal, front-end, detector, and readout impairments are represented by the effective output-noise term $w_i[k, m]$. Because each subslot averages many RF noise samples, $w_i[k, m]$ is approximated as independent zero-mean Gaussian noise with constant variance. An exact pre-detection Gaussian model would instead yield signal-dependent noncentral chi-squared observations and is not considered here.

Let $u_i(t) = [\mathbf{H}\mathbf{x}(t)]_i + b_i$. Squaring the corresponding real RF waveform gives $[\sqrt{2} \text{Re}\{u_i(\tau) e^{j2\pi f_c \tau}\}]^2 = |u_i(\tau)|^2 + \text{Re}\{u_i^2(\tau) e^{j4\pi f_c \tau}\}$. Since $u_i(t)$ varies on the envelope time scale, the $2f_c$ term averages out when $\Delta_t f_c \gg 1$. The remaining envelope energy contains the phase-sensitive term $2 \text{Re}\{b_i^* [\mathbf{H}\mathbf{x}(t)]_i\}$. Thus, (9) represents RF energy integration rather than coherent downconversion or I/Q sampling. If the envelope is approximately constant within a subslot, stacking the N_r outputs yields

$$\mathbf{z}[k, m] = |\mathbf{H}\mathbf{x}[k, m] + \mathbf{b}|^2 + \mathbf{w}[k, m] \in \mathbb{R}^{N_r}, \quad (10)$$

where $\mathbf{x}[k, m] = \mathbf{x}(t_{k,m})$ and the squared modulus is element-wise. The MN_r samples associated with each symbol are retained as a vector. Spatial diversity is supplied by the receiving units and their reference phases, whereas $M > 1$ is used only when temporal envelope samples are needed; it is not the number of reference phases.

To relate these energy samples to the transmitted symbols, define $\mathbf{D}_P = \text{diag}(\sqrt{P_1}, \dots, \sqrt{P_L})$, $[\mathbf{P}]_{kM+m,j} = p(t_{k,m} - jT_s)$, and stack $\mathbf{s} = [\mathbf{s}_1^T, \dots, \mathbf{s}_L^T]^T$. The matrix $\mathbf{P}_{\text{tx}} \triangleq \mathbf{D}_P \otimes$

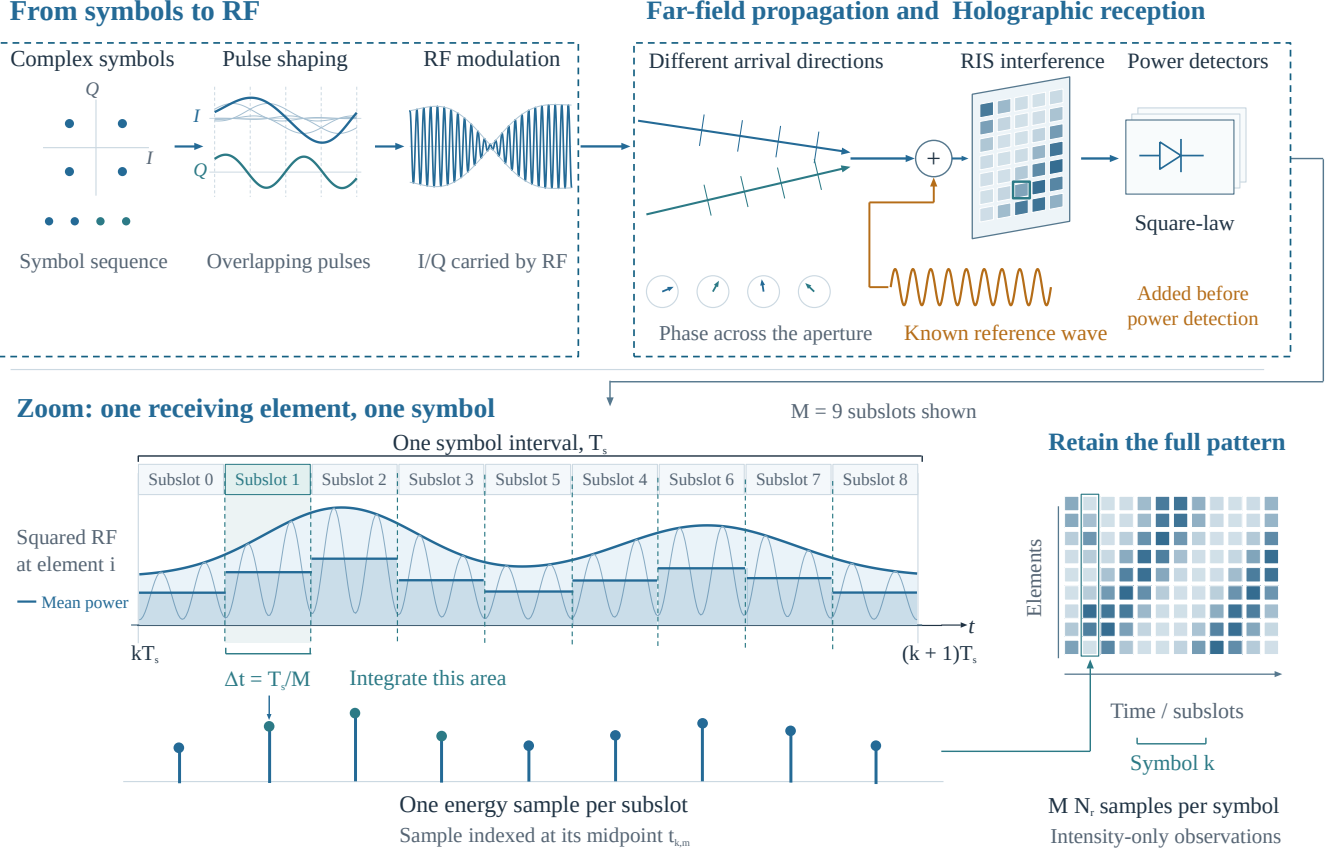


Fig. 2. Reference-assisted holographic reception model. Finite-alphabet symbols are pulse-shaped and upconverted to RF signals. After far-field propagation, the incident fields from different users form direction-dependent phase patterns across the holographic aperture. A known reference wave is added at the RF front end before square-law detection, converting relative phase information into intensity variations. When waveform variation within a symbol must be resolved, the symbol interval is divided into M subslots; otherwise, $M = 1$. The receiver preserves the resulting spatial intensity pattern for channel estimation and symbol detection.

$\mathbf{P} \in \mathbb{C}^{LKM \times LK}$ maps $\mathbf{s}$ to the sampled user envelopes, where $\otimes$ denotes the Kronecker product.

For general pulse shaping, neighboring symbols may overlap, so the KMN_r measurements must be processed jointly. Let Π reorder the user-major envelope $\mathbf{P}_{\text{tx}}\mathbf{s}$ into time-major form. Stacking (10) over all subslots then yields the compact block-level model

$$\mathbf{z} = |\mathbf{G}\mathbf{s} + \mathbf{b}_{\text{eq}}|^2 + \mathbf{w}, \quad (11)$$

where $\mathbf{G} \triangleq (\mathbf{I}_{KM} \otimes \mathbf{H})\Pi\mathbf{P}_{\text{tx}} \in \mathbb{C}^{KMN_r \times LK}$ and $\mathbf{b}_{\text{eq}} \triangleq \mathbf{1}_{KM} \otimes \mathbf{b}$. Thus, $\mathbf{s} \in \mathcal{S}^{LK}$ and $\mathbf{z}, \mathbf{w} \in \mathbb{R}^{KMN_r}$. When intersymbol overlap is negligible, (11) separates into symbol-wise models; otherwise, it describes joint block detection.

D. Two-Phase Transmission Protocol and Unified Observation Model

The receiver uses a training phase to estimate the channel from known pilots and a data phase to detect finite-alphabet symbols using the estimated channel. Both phases follow (11), but their unknowns have different structures.

1) *Training Phase*: Let $\mathbf{x}_p[t] \in \mathbb{C}^L$ be the known waveform in pilot subslot $t = 0, \dots, T_p - 1$, and define $\mathbf{h} = \text{vec}(\mathbf{H})$.

Stacking the element-wise observations and using $\mathbf{H}\mathbf{x}_p[t] = (\mathbf{x}_p^T[t] \otimes \mathbf{I}_{N_r})\mathbf{h}$ gives

$$\mathbf{z}_p = |\mathbf{G}_p\mathbf{h} + \mathbf{b}_p|^2 + \mathbf{w}_p, \quad (12)$$

where $\mathbf{G}_p$ stacks $\mathbf{x}_p^T[t] \otimes \mathbf{I}_{N_r}$, and $\mathbf{b}_p$ and $\mathbf{w}_p$ stack the reference and noise. For the fixed spatial code considered here, $\mathbf{b}_p$ repeats the same calibrated N_r -entry phase map at every pilot time. The far-field structure in (6) will later reduce the unknown from the full vector $\mathbf{h}$ to the DOAs and complex path gains.

2) *Data Transmission Phase*: For a data block of K_d symbols per user, let $\mathbf{s}_d \in \mathcal{S}^{LK_d}$ denote the stacked symbol vector. Denote the true noiseless response by $\boldsymbol{\mu}_d(\mathbf{s}; \mathbf{H}) \triangleq |\mathbf{G}_d(\mathbf{H})\mathbf{s} + \mathbf{b}_d|^2$. Using $\hat{\mathbf{H}}$, the receiver forms the detection template $\hat{\boldsymbol{\mu}}_d(\mathbf{s}) \triangleq |\mathbf{G}_d(\hat{\mathbf{H}})\mathbf{s} + \hat{\mathbf{b}}_d|^2$. The actual observation can therefore be written as

$$\mathbf{z}_d = \hat{\boldsymbol{\mu}}_d(\mathbf{s}_d) + \mathbf{e}_{\text{CSI}}(\mathbf{s}_d) + \mathbf{w}_d, \quad (13)$$

where $\mathbf{e}_{\text{CSI}}(\mathbf{s}) \triangleq \boldsymbol{\mu}_d(\mathbf{s}; \mathbf{H}) - \hat{\boldsymbol{\mu}}_d(\mathbf{s})$ is the generally symbol-dependent mismatch error caused by imperfect channel state information (CSI) and vanishes for perfect CSI. Here, $Q_d = N_rMK_d$, and $\hat{\mathbf{b}}_d = \mathbf{b}_d$ is the calibrated receiver reference.

Thus, the training and data phases share the same affine intensity structure, but the latter performs finite-alphabet detection with an estimated observation map.

III. UNIQUENESS AND STABILITY OF REFERENCE-ASSISTED ENERGY DETECTION

Reliable recovery from intensity-only measurements requires more than the existence of a solution: distinct channels or symbol vectors must produce distinct energy observations, and small measurement perturbations must not cause arbitrarily large reconstruction errors. These requirements are captured by uniqueness and stability, respectively, and are fundamental to determining whether the reference wave and spatial sampling provide sufficient information. Exploiting the quadratic structure of the reference-assisted energy mapping, we establish an exact secant-based uniqueness condition and derive lower and upper Lipschitz bounds that quantify robustness. The resulting framework applies to both continuous channel estimation and finite-alphabet symbol detection.

A. Unified Affine Phase-Retrieval Model

The training- and data-phase observations derived in Section II share the form

$$\mathbf{z} = \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}) + \mathbf{w}, \quad \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}) \triangleq |\mathbf{G}\boldsymbol{\theta} + \mathbf{b}|^2, \quad (14)$$

where $\boldsymbol{\theta} \in \mathcal{X} \subseteq \mathbb{C}^D$ denotes the unknown complex-valued parameter vector, $\mathbf{G} \in \mathbb{C}^{Q \times D}$ is the measurement matrix, $\mathbf{b} \in \mathbb{C}^Q$ is the known reference vector, and $\mathbf{z}, \mathbf{w} \in \mathbb{R}^Q$ denote the intensity observation and noise vectors, respectively. Here, D is the complex dimension of the unknown parameter vector, while Q denotes the total number of scalar intensity measurements. The set $\mathcal{X}$ specifies the feasible domain of $\boldsymbol{\theta}$, and both the modulus and squaring operations are applied element-wise. Thus, $\mathcal{T}_{\mathbf{G},\mathbf{b}}$ is the affine magnitude-squared mapping associated with affine phase retrieval [40]. For full-channel estimation in the training phase, $\boldsymbol{\theta} = \mathbf{h} = \text{vec}(\mathbf{H})$, $Q = N_r T_p$, and $D = N_r L$. In the data phase, $\boldsymbol{\theta} = \mathbf{s}_d \in \mathcal{S}^{LK_d}$, $Q = Q_d$, and $D = LK_d$.

The reference changes the ambiguity class fundamentally. If $\mathbf{b} = 0$, $\mathcal{T}_{\mathbf{G},0}(e^{j\varphi}\boldsymbol{\theta}) = \mathcal{T}_{\mathbf{G},0}(\boldsymbol{\theta})$, so uniqueness is defined only modulo a global phase. If $\mathbf{b} \neq 0$, the expansion

$$\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}) = |\mathbf{G}\boldsymbol{\theta}|^2 + |\mathbf{b}|^2 + 2 \text{Re}\{\mathbf{b}^* \odot \mathbf{G}\boldsymbol{\theta}\} \quad (15)$$

contains phase-sensitive interference, so absolute uniqueness on $\mathcal{X}$ becomes possible. A nonzero reference is not sufficient by itself, however: its phases must probe every feasible field difference in nondegenerate real directions.

For $\boldsymbol{\theta} \in \mathbb{C}^D$, define the realification $\boldsymbol{\theta}_{\mathbb{R}} = [\text{Re}\{\boldsymbol{\theta}\}^T, \text{Im}\{\boldsymbol{\theta}\}^T]^T \in \mathbb{R}^{2D}$. For two distinct parameters, define the energy-domain distance

$$d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2) \triangleq \|\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_1) - \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_2)\|_2. \quad (16)$$

The optimal lower and upper stability bounds on $\mathcal{X}$ are

$$\alpha_{\mathcal{X}} \triangleq \inf_{\substack{\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X} \\ \boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2}} \frac{d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2)}{\|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2}, \quad \beta_{\mathcal{X}} \triangleq \sup_{\substack{\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X} \\ \boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2}} \frac{d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2)}{\|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2}. \quad (17)$$

Thus, the inverse problem is globally stable on $\mathcal{X}$ if $0 < \alpha_{\mathcal{X}} \leq \beta_{\mathcal{X}} < \infty$. The lower bound is the important one: it excludes pairs that are far apart in parameter space but arbitrarily close in energy space.

B. Exact Secant Identity and Global Uniqueness

The quadratic structure of (14) gives an exact identity, which is stronger than a first-order approximation. For a point $\boldsymbol{\vartheta} \in \mathbb{C}^D$, define the real Jacobian

$$\mathbf{J}(\boldsymbol{\vartheta}) \triangleq 2[\text{Re}\{\text{diag}[(\mathbf{G}\boldsymbol{\vartheta} + \mathbf{b})^*]\mathbf{G}\}, \\ -\text{Im}\{\text{diag}[(\mathbf{G}\boldsymbol{\vartheta} + \mathbf{b})^*]\mathbf{G}\}] \in \mathbb{R}^{Q \times 2D}. \quad (18)$$

Theorem 1 (Exact secant identity and pairwise uniqueness). *For any $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathbb{C}^D$, let $\boldsymbol{\mu} = (\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2)/2$ and $\boldsymbol{\delta} = \boldsymbol{\theta}_1 - \boldsymbol{\theta}_2$. Then*

$$\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_1) - \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_2) = \mathbf{J}(\boldsymbol{\mu})\boldsymbol{\delta}_{\mathbb{R}}. \quad (19)$$

Consequently, the mapping $\mathcal{T}_{\mathbf{G},\mathbf{b}}$ is injective on $\mathcal{X}$ if and only if, for every distinct $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X}$,

$$\mathbf{J}\left(\frac{\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2}{2}\right)(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)_{\mathbb{R}} \neq \mathbf{0}. \quad (20)$$

Proof. Applying $|a|^2 - |c|^2 = \text{Re}\{(a+c)^*(a-c)\}$ element-wise gives $\text{Re}\{[\mathbf{G}(\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2) + 2\mathbf{b}]^* \odot \mathbf{G}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)\}$, which equals (19) after realification. Hence, two feasible points have identical intensity observations exactly when the right-hand side of (19) vanishes. Excluding this possibility for every distinct pair is therefore equivalent to injectivity on $\mathcal{X}$, proving (20). $\square$

For a full-dimensional continuous set, $\text{rank}(\mathbf{G}) = D$ is necessary but not sufficient: a nonzero field difference can still vanish after the reference-dependent real projection. A sharper conclusion is available for finite hypotheses.

Corollary 1 (Generic-reference uniqueness for finite hypotheses). *Let $\Xi = \{\xi_1, \dots, \xi_M\}$ be finite, with pre-reference field codewords $\mathbf{u}_i = \mathbf{f}(\xi_i) \in \mathbb{C}^Q$. For $i \neq j$, set $\mathbf{d}_{ij} = \mathbf{u}_i - \mathbf{u}_j$ and $\mathbf{a}_{ij} = (|\mathbf{u}_i|^2 - |\mathbf{u}_j|^2)/2$. A pair with $\mathbf{d}_{ij} = \mathbf{0}$ is indistinguishable for every reference; otherwise, its ambiguous references form the proper real affine set*

$$\mathcal{B}_{ij}^{\text{bad}} = \{\mathbf{b} \in \mathbb{C}^Q : \text{Re}\{\mathbf{b}^* \odot \mathbf{d}_{ij}\} = -\mathbf{a}_{ij}\}. \quad (21)$$

If all field codewords are distinct, the finite union $\bigcup_{i < j} \mathcal{B}_{ij}^{\text{bad}}$ has measure zero in $\mathbb{C}^Q$; hence, an absolutely continuously distributed reference makes $|\mathbf{u}_i + \mathbf{b}|^2$ pairwise distinct with probability one.

Proof. For two hypotheses in the affine field model, let $\boldsymbol{\mu}_{ij} = (\boldsymbol{\theta}_i + \boldsymbol{\theta}_j)/2$ and $\boldsymbol{\delta}_{ij} = \boldsymbol{\theta}_i - \boldsymbol{\theta}_j$, so that $\mathbf{u}_i = \mathbf{G}\boldsymbol{\theta}_i$, $\mathbf{u}_j = \mathbf{G}\boldsymbol{\theta}_j$, and $\mathbf{d}_{ij} = \mathbf{G}\boldsymbol{\delta}_{ij}$. Expanding the Jacobian-secant product in (20) gives

$$\begin{aligned} & \mathbf{J}(\boldsymbol{\mu}_{ij})(\boldsymbol{\delta}_{ij})_{\mathbb{R}} \\ &= 2 \text{Re}\left\{\left(\frac{\mathbf{u}_i + \mathbf{u}_j}{2} + \mathbf{b}\right)^* \odot (\mathbf{u}_i - \mathbf{u}_j)\right\} \\ &= \text{Re}\{(\mathbf{u}_i + \mathbf{u}_j)^* \odot \mathbf{d}_{ij}\} + 2 \text{Re}\{\mathbf{b}^* \odot \mathbf{d}_{ij}\} \\ &= |\mathbf{u}_i|^2 - |\mathbf{u}_j|^2 + 2 \text{Re}\{\mathbf{b}^* \odot \mathbf{d}_{ij}\} \\ &= 2(\mathbf{a}_{ij} + \text{Re}\{\mathbf{b}^* \odot \mathbf{d}_{ij}\}). \end{aligned}$$

The cross terms in the third line cancel after taking the real part. Hence, the pair is ambiguous precisely when $\text{Re}\{\mathbf{b}^* \odot \mathbf{d}_{ij}\} = -\mathbf{a}_{ij}$, which proves (21). This final identity depends only on the two field codewords and therefore also holds for a general finite mapping $\mathbf{f}$.

If $d_{ij,q} = 0$, then $u_{i,q} = u_{j,q}$ and $a_{ij,q} = 0$, so that coordinate imposes no restriction. If $d_{ij,q} \neq 0$, all ambiguous reference values are

$$b_q = \left(-\frac{a_{ij,q}}{|d_{ij,q}|^2} + j\tau_q \right) d_{ij,q}, \quad \tau_q \in \mathbb{R},$$

which is one real affine constraint in $\mathbb{C}$. Therefore, if $r_{ij} = |\{q : d_{ij,q} \neq 0\}| \geq 1$, then $\mathcal{B}_{ij}^{\text{bad}}$ has real dimension $2Q - r_{ij} < 2Q$ and hence Lebesgue measure zero. A finite union of such sets remains measure zero, establishing the almost-sure uniqueness claim. $\square$

For $b_q = \rho_q e^{j\chi_q}$, an active coordinate is ambiguous only when $\rho_q |d_{ij,q}| \cos(\angle d_{ij,q} - \chi_q) = -a_{ij,q}$, which excludes at most two phases. Hence, if the reference support intersects every nonzero field difference, independently and continuously varying phases remove finite-set ambiguities almost surely. This argument does not directly apply to fixed or spatially coupled finite-phase codes, whose validity must instead be verified through (21) or the resulting minimum distance. For $\mathbf{f}(\boldsymbol{\theta}) = \mathbf{G}\boldsymbol{\theta}$, the required condition becomes $\mathbf{G}(\boldsymbol{\theta}_i - \boldsymbol{\theta}_j) \neq \mathbf{0}$. Thus, full column rank is sufficient but not necessary on a finite set, whereas any difference in $\text{Null}(\mathbf{G})$ remains unrecoverable for all references. The same reasoning applies to a discretized parametric set $\mathbf{u}_i = \mathbf{g}_p(\boldsymbol{\eta}_i)$; continuous nuisance parameters are treated separately below.

Generic injectivity on a finite set does not by itself ensure a useful noise margin. Returning to a general feasible set, let $\mathcal{M}_{\mathcal{X}} = \{(\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2)/2 : \boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X}\}$ be the midpoint set. The exact identity yields computable two-sided bounds.

Theorem 2 (Jacobian stability bounds). *Define*

$$\underline{\sigma}_J \triangleq \inf_{\boldsymbol{\mu} \in \mathcal{M}_{\mathcal{X}}} \sigma_{\min}(\mathbf{J}(\boldsymbol{\mu})), \quad \overline{\sigma}_J \triangleq \sup_{\boldsymbol{\mu} \in \mathcal{M}_{\mathcal{X}}} \sigma_{\max}(\mathbf{J}(\boldsymbol{\mu})). \quad (22)$$

Then

$$\underline{\sigma}_J \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2 \leq d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2) \leq \overline{\sigma}_J \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2 \quad (23)$$

for all pairs in $\mathcal{X}$. Consequently, $\alpha_{\mathcal{X}} \geq \underline{\sigma}_J$ and $\beta_{\mathcal{X}} \leq \overline{\sigma}_J$.

Proof. Apply the singular-value inequalities to (19) over the midpoint set. $\square$

The bound $\underline{\sigma}_J > 0$ is a convenient sufficient condition for global stability, but it can be conservative because it tests every direction at every midpoint, including directions that may not connect two feasible points. The exact constant in (??), or equivalently the restricted secants in (19), is the sharp criterion.

The stability constants also depend on the balance between the received field and the reference wave. To make this dependence explicit, let $u_q(\boldsymbol{\theta}) = [\mathbf{G}\boldsymbol{\theta}]_q$ and, whenever $u_q(\boldsymbol{\theta}) \neq 0$, define the per-measurement reference-to-signal energy ratio and the corresponding normalized interference depth as

$$\gamma_q(\boldsymbol{\theta}) \triangleq \frac{|b_q|^2}{|u_q(\boldsymbol{\theta})|^2}, \quad \kappa_q(\boldsymbol{\theta}) \triangleq \frac{2\sqrt{\gamma_q(\boldsymbol{\theta})}}{1 + \gamma_q(\boldsymbol{\theta})}. \quad (24)$$

Writing $\Delta\phi_q = \angle u_q - \angle b_q$, the q th noiseless measurement becomes

$$[\mathcal{T}_{\mathbf{G}, \mathbf{b}}(\boldsymbol{\theta})]_q = (|u_q|^2 + |b_q|^2) [1 + \kappa_q(\boldsymbol{\theta}) \cos(\Delta\phi_q)]. \quad (25)$$

Here $0 \leq \kappa_q \leq 1$, with its maximum attained when the two incident fields have equal energy. It approaches zero when either field dominates, showing that the phase-sensitive interference then occupies only a small fraction of the detected energy.

C. Parametric Channel Uniqueness Requirements

The affine phase-retrieval result in [40, Theorem 3.1] states that a mapping on $\mathbb{C}^D$ is globally injective if and only if its real Jacobian has rank $2D$ at every point. For unstructured training, $D_h = N_r L$ and $Q_p = N_r T_p$; the universal measurement result in [40, Theorem 3.2] therefore implies the necessary global count $T_p \geq 3L$. This result is only a benchmark for the structured pilot-reference map used here.

The far-field channel instead lies on a real parameter manifold of dimension $p = (d_a + 2)L$, where d_a is the number of angular parameters per user. With $\mathcal{T}_p(\boldsymbol{\eta}) = |\mathbf{G}_p \mathbf{h}(\boldsymbol{\eta}) + \mathbf{b}_p|^2$, define $\mathbf{J}_{\boldsymbol{\eta}} = \partial \mathcal{T}_p / \partial \boldsymbol{\eta}^T$. At an interior regular point, local identifiability requires $\text{rank}(\mathbf{J}_{\boldsymbol{\eta}}) = p$, and hence $N_r M K_p \geq p$ when $T_p = M K_p$. Spatial measurements can therefore reduce temporal training only because the far-field parameter dimension does not scale with N_r .

For the unstructured model, $\mathbf{G}_p = \mathbf{X}_p \otimes \mathbf{I}_{N_r}$, so $\text{rank}(\mathbf{G}_p) = N_r \text{rank}(\mathbf{X}_p)$ and $\sigma_{\min}(\mathbf{G}_p) = \sigma_{\min}(\mathbf{X}_p)$. The resulting necessary hierarchy is

$$\begin{aligned} T_p &\geq L && \text{for complex-field preservation,} \\ T_p &\geq 2L && \text{for regular local intensity recovery,} \\ T_p &\geq 3L && \text{necessary for global recovery.} \end{aligned} \quad (26)$$

These counts are necessary but not sufficient. Full rank also requires nonzero path gains, an ambiguity-free array geometry, and pilot and reference phases that preserve all relevant tangent directions. Global uniqueness further requires the pairwise condition in (20), while the magnitude of $\sigma_{\min}(\mathbf{J}_{\boldsymbol{\eta}})$ determines the local noise sensitivity.

D. Finite-Alphabet Uniqueness and Noise Margin

For data detection, let $\mathcal{X}_d = \mathcal{S}^{L K_d}$. Since this candidate set is finite, Corollary 1 shows that noiseless uniqueness is equivalent to a positive minimum separation, defined as

$$d_{\min, d}^{(E)} \triangleq \min_{\substack{\mathbf{s}_1, \mathbf{s}_2 \in \mathcal{X}_d \\ \mathbf{s}_1 \neq \mathbf{s}_2}} \|\mathbf{G}_d \mathbf{s}_1 + \mathbf{b}_d\|^2 - \|\mathbf{G}_d \mathbf{s}_2 + \mathbf{b}_d\|^2. \quad (27)$$

Theorem 3 (Noise-limited finite-alphabet reliability). *The finite-alphabet symbols are uniquely identifiable in the noiseless case if and only if $d_{\min, d}^{(E)} > 0$. For an arbitrary additive perturbation, the nearest-intensity detector returns the transmitted vector whenever*

$$\|\mathbf{w}\|_2 < \frac{d_{\min, d}^{(E)}}{2}. \quad (28)$$

Under exact CSI and the adopted white Gaussian effective output-noise model, $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \sigma_w^2 \mathbf{I})$, the error probability of

the binary ML comparison between distinct candidates $\mathbf{s}_i$ and $\mathbf{s}_j$ is

$$P_{i \rightarrow j} = Q\left(\frac{d_E(\mathbf{s}_i, \mathbf{s}_j)}{2\sigma_w}\right), \quad (29)$$

where $Q(x) \triangleq (2\pi)^{-1/2} \int_x^\infty e^{-t^2/2} dt$ is the standard Gaussian tail function. If $M_c = |\mathcal{X}_d| = |\mathcal{S}|^{LK_d}$ is the number of candidate vectors, the block error probability satisfies

$$P_B \leq (M_c - 1)Q\left(\frac{d_{\min,d}^{(E)}}{2\sigma_w}\right). \quad (30)$$

Proof. Let $\mu_k = |\mathbf{G}_d \mathbf{s}_k + \mathbf{b}_d|^2$. The triangle inequality gives the decoding radius in (28). If $\mathbf{s}_i$ is transmitted, the binary ML comparison favors $\mathbf{s}_j$ only if $\mathbf{w}^T(\mu_j - \mu_i) \geq d_E^2(\mathbf{s}_i, \mathbf{s}_j)/2$. The left-hand side is Gaussian with variance $\sigma_w^2 d_E^2(\mathbf{s}_i, \mathbf{s}_j)$, which gives (29). Applying the union bound over the remaining $M_c - 1$ candidates and using $d_E(\mathbf{s}_i, \mathbf{s}_j) \geq d_{\min,d}^{(E)}$ gives (30). $\square$

Thus, $d_{\min,d}^{(E)}$ provides a common measure of finite-alphabet identifiability and noise robustness: positivity guarantees noiseless uniqueness, half of the distance is a deterministic decoding radius, and its value directly controls the Gaussian error bounds. Because it depends jointly on $\mathbf{G}_d$, $\mathbf{b}_d$, and the constellation, it also serves as the reference-design metric and distance certificate used in Section IV.

E. Spatial Four-Phase Coding and Energy Allocation

The preceding conditions apply to any calibrated reference map. We now examine the parallel hardware architecture in which each spatial coordinate is assigned one of four phases,

$$b_q = \rho_q e^{j\chi_q}, \quad \chi_q \in \{0, \pi/2, \pi, 3\pi/2\}, \quad \rho_q > 0. \quad (31)$$

Unlike temporal phase shifting, the four observations do not correspond to repeated measurements of the same unknown field sample. A useful guarantee is therefore obtained over the ensemble of balanced spatial phase assignments.

Theorem 4 (Mean separation of a spatial four-phase code). *Let each phase in (31) have a uniform marginal distribution over the four states, independently of a fixed pair θ_1, θ_2 . Define $\mathbf{u}_i = \mathbf{G}\theta_i$, $\mathbf{c} = |\mathbf{u}_1|^2 - |\mathbf{u}_2|^2$, and $\Delta\mathbf{u} = \mathbf{u}_1 - \mathbf{u}_2$. Then*

$$\mathbb{E}_{\mathcal{X}}[\|\mathcal{T}_{\mathbf{G},\mathbf{b}}(\theta_1) - \mathcal{T}_{\mathbf{G},\mathbf{b}}(\theta_2)\|_2^2] = \|\mathbf{c}\|_2^2 + 2 \sum_q \rho_q^2 |\Delta u_q|^2. \quad (32)$$

Consequently, with $\rho_{\min} = \min_q \rho_q$,

$$\mathbb{E}_{\mathcal{X}}[\|\Delta\mu\|_2^2] \geq 2\rho_{\min}^2 \|\mathbf{G}(\theta_1 - \theta_2)\|_2^2. \quad (33)$$

The assumption includes a uniformly randomized balanced assignment, for which the four states occur equally often but the unit phases need not be mutually independent.

Proof. For coordinate q , the intensity difference is $\Delta\mu_q = c_q + 2\rho_q \operatorname{Re}\{e^{-j\chi_q} \Delta u_q\}$. The four-state average satisfies $\mathbb{E}_{\chi_q}[\operatorname{Re}\{e^{-j\chi_q} \Delta u_q\}] = 0$ and $\mathbb{E}_{\chi_q}[\operatorname{Re}^2\{e^{-j\chi_q} \Delta u_q\}] = |\Delta u_q|^2/2$. Hence, $\mathbb{E}_{\chi_q}[\Delta\mu_q^2] = c_q^2 + 2\rho_q^2 |\Delta u_q|^2$. Summing over the spatial coordinates proves (32), and lower bounding each ρ_q by $\rho_{\min}$ gives (33). $\square$

The theorem shows why balanced spatial phase coding is useful on average, but it is not a deterministic certificate for every realized phase map. A fixed hardware code must still be validated through its actual Jacobian, restricted secant constant, or finite-alphabet minimum distance. The same averaged identity also gives a closed-form energy allocation for each candidate pair.

Corollary 2 (Pairwise energy allocation under a fixed incident-energy budget). *Under the noise model of Theorem 3, suppose that the candidate fields are $\mathbf{u}_i = \mathbf{G}_d \mathbf{s}_i$. Define the average received-field energy and reference energy per measurement as*

$$P_s \triangleq \frac{1}{M_c Q_d} \sum_{i=1}^{M_c} \|\mathbf{u}_i\|_2^2, \quad P_b \triangleq \frac{1}{Q_d} \|\mathbf{b}_d\|_2^2, \quad (34)$$

and let $P_T = P_s + P_b$. Writing $\mathbf{u}_i = \sqrt{P_s} \mathbf{v}_i$ gives $(M_c Q_d)^{-1} \sum_i \|\mathbf{v}_i\|_2^2 = 1$. Suppose that P_T is fixed and the spatial phases are drawn uniformly from the four states in (31), with $|b_q|^2 = P_b$. Then

$$\begin{aligned} \mathbb{E}_{\mathcal{X}}[d_{E,ij}^2] &= P_s^2 A_{ij} + 2P_s P_b B_{ij}, \\ A_{ij} &= \|\mathbf{v}_i\|^2 - \|\mathbf{v}_j\|^2, \\ B_{ij} &= \|\mathbf{v}_i - \mathbf{v}_j\|_2^2. \end{aligned} \quad (35)$$

For $0 \leq A_{ij} < B_{ij}$, this ensemble-mean squared distance is maximized by

$$P_s^* = \frac{B_{ij}}{2B_{ij} - A_{ij}} P_T, \quad \frac{P_b^*}{P_s^*} = 1 - \frac{A_{ij}}{B_{ij}}. \quad (36)$$

In particular, phase-only pairs satisfy $A_{ij} = 0$ and are optimally separated by $P_s = P_b = P_T/2$. Therefore, when the minimum-distance pairs are phase dominated, $A_{ij} \ll B_{ij}$, equal signal and reference energies provide a useful nominal operating point rather than a universal optimum. For amplitude-dominated pairs in higher-order QAM, $A_{ij} \geq B_{ij}$ may instead place the pairwise optimum at the boundary. Moreover, the active minimum-distance pair can change with the power split, so the system-level allocation must be assessed using the actual $d_{\min,d}^{(E)}$.

Proof. For measurement coordinate q and reference phase χ_q , let $a_q = |v_{i,q}|^2 - |v_{j,q}|^2$ and $\delta v_q = v_{i,q} - v_{j,q}$. The noiseless intensity difference is

$$\Delta\mu_q = P_s a_q + 2\sqrt{P_s P_b} \operatorname{Re}\{e^{-j\chi_q} \delta v_q\}. \quad (37)$$

Taking the four-state expectation cancels the cross term and gives a mean squared projection of $|\delta v_q|^2/2$. Summing over q gives (35). Substituting $P_b = P_T - P_s$ and differentiating yields (36); its second derivative is negative for $A_{ij} < B_{ij}$. When $A_{ij} = 0$,

$$\mathbb{E}_{\mathcal{X}}[d_{E,ij}^2] = 2B_{ij} P_s (P_T - P_s) \leq \frac{B_{ij} P_T^2}{2}, \quad (38)$$

with equality at equal energy. Thus, the allocation maximizes the average geometric separation of that pair over the spatial-code ensemble. The error probability for a realized code is still determined by its actual $d_{E,ij}$ through (29). $\square$

The finite optimum in Corollary 2 relies on a fixed total-energy budget. If P_s and σ_w^2 are fixed while only the reference power is increased, the ideal model does not generally predict an equal-energy optimum. In practice, the locally generated reference can be adjusted using the average received-field energy estimated from pilots or spatial averages, so that severe energy imbalance is avoided subject to the detector dynamic range; the preferred ratio remains scenario dependent.

Taken together, the analysis identifies the local conditioning of the parametric training Jacobian and the finite-alphabet minimum energy distance as useful reference-design metrics. They are used here to guide and evaluate feasible reference configurations; global optimization over hardware-constrained phase maps is not pursued.

IV. ESTIMATION AND DETECTION ALGORITHMS

Guided by Section III, the continuous, low-dimensional channel parameters are refined by damped Gauss–Newton (GN) where the intensity Jacobian is well conditioned. With the channel fixed, the data symbols form a finite set governed by the energy-domain minimum distance; maximum-likelihood (ML) detection is the benchmark and a reference-assisted Gerchberg–Saxton (GS) iteration provides the scalable implementation.

A. Overall Two-Stage Recovery Framework

The proposed receiver retains the two-phase structure of Section II. During training, the far-field model in (6) is parameterized by the real vector

$$\boldsymbol{\eta} = [\boldsymbol{\theta}^T \quad \boldsymbol{\phi}^T \quad \text{Re}\{\boldsymbol{\alpha}\}^T \quad \text{Im}\{\boldsymbol{\alpha}\}^T]^T \in \mathbb{R}^{D_\eta}, \quad (39)$$

where $\boldsymbol{\theta}$, $\boldsymbol{\phi}$, and $\boldsymbol{\alpha}$ collect the polar angles, azimuth angles, and complex path gains, respectively; thus, $D_\eta = 4L$. Let $\mathbf{g}_p(\boldsymbol{\eta}) = \text{col}_t\{\mathbf{H}(\boldsymbol{\eta})\mathbf{x}_p[t]\} \in \mathbb{C}^{Q_p}$, where $Q_p = N_r T_p$. The parametric training model is

$$\mathbf{z}_p = |\mathbf{g}_p(\boldsymbol{\eta}) + \mathbf{b}_p|^2 + \mathbf{w}_p, \quad (40)$$

After Gauss–Newton refinement, the receiver reconstructs $\hat{\mathbf{H}} = \mathbf{H}(\hat{\boldsymbol{\eta}})$ and forms $\hat{\mathbf{G}}_d = \mathbf{G}_d(\hat{\mathbf{H}})$ in (13). The training reference is designed to improve the conditioning of $\mathbf{J}_p$, whereas the data reference is designed to enlarge the finite-alphabet energy distance. The following two subsections give the corresponding estimators.

B. Joint DOA and Channel-Gain Estimation

1) *Nonlinear Least-Squares Formulation*: Based on (40), joint DOA and channel-gain estimation is formulated as

$$\hat{\boldsymbol{\eta}} = \arg \min_{\boldsymbol{\eta} \in \Omega} F_p(\boldsymbol{\eta}), \quad F_p(\boldsymbol{\eta}) \triangleq \frac{1}{2} \left\| \mathbf{z}_p - |\mathbf{g}_p(\boldsymbol{\eta}) + \mathbf{b}_p|^2 \right\|_2^2, \quad (41)$$

where $\Omega = \Theta^L \times \Phi^L \times \mathcal{A}^L$ is a compact feasible region. Here, $\Theta \subseteq [0, \pi/2]$ and $\Phi \subseteq [-\pi/2, \pi/2]$ specify the visible angular sector, while $\mathcal{A} = \{\alpha \in \mathbb{C} : |\alpha| \leq \alpha_{\max}\}$ imposes a loose gain bound set by the receiver normalization or automatic gain control (AGC) range. No positive lower gain bound is imposed. This physical parameterization uses $4L$ real

unknowns rather than the $2N_r L$ real entries of an unstructured channel. Its Jacobian also exposes weak gains and poorly separated DOAs through small singular values.

2) *Jacobian of the Parametric Intensity Mapping*: At the t -th pilot instant, the noise-free received field is given by

$$\mathbf{g}_p[t] = \sum_{l=1}^L \alpha_l x_{p,l}[t] \mathbf{v}_l, \quad (42)$$

where $x_{p,l}[t]$ denotes the pilot symbol transmitted by the l -th user, corresponding to the l -th entry of the pilot vector $\mathbf{x}_p[t]$.

Let $\boldsymbol{\psi}_l \in \mathbb{R}^{N_r}$ denote the phase vector whose $\iota(n, m)$ -th entry is $[\boldsymbol{\psi}_l]_{\iota(n, m)} = \psi_{n, m}^{(\iota)}$. Since $\mathbf{v}_l(\theta_l, \phi_l) = \exp(j\boldsymbol{\psi}_l)$, where the exponential is applied element-wise, its angular derivatives can be written as

$$\frac{\partial \mathbf{v}_l}{\partial \theta_l} = j \frac{\partial \boldsymbol{\psi}_l}{\partial \theta_l} \odot \mathbf{v}_l, \quad \frac{\partial \mathbf{v}_l}{\partial \phi_l} = j \frac{\partial \boldsymbol{\psi}_l}{\partial \phi_l} \odot \mathbf{v}_l, \quad (43)$$

where the $\iota(n, m)$ -th entries are $[\partial \boldsymbol{\psi}_l / \partial \theta_l]_{\iota(n, m)} = k_0 \cos \theta_l (y_n \cos \phi_l + z_m \sin \phi_l)$ and $[\partial \boldsymbol{\psi}_l / \partial \phi_l]_{\iota(n, m)} = k_0 \sin \theta_l (-y_n \sin \phi_l + z_m \cos \phi_l)$.

Using (42), and defining $\alpha_{l,R} = \text{Re}\{\alpha_l\}$ and $\alpha_{l,I} = \text{Im}\{\alpha_l\}$, the field derivatives at pilot instant t are

$$\begin{aligned} \frac{\partial \mathbf{g}_p[t]}{\partial \theta_l} &= \alpha_l x_{p,l}[t] \frac{\partial \mathbf{v}_l}{\partial \theta_l}, & \frac{\partial \mathbf{g}_p[t]}{\partial \phi_l} &= \alpha_l x_{p,l}[t] \frac{\partial \mathbf{v}_l}{\partial \phi_l}, \\ \frac{\partial \mathbf{g}_p[t]}{\partial \alpha_{l,R}} &= x_{p,l}[t] \mathbf{v}_l, & \frac{\partial \mathbf{g}_p[t]}{\partial \alpha_{l,I}} &= j x_{p,l}[t] \mathbf{v}_l. \end{aligned} \quad (44)$$

The derivatives over all pilot instants are stacked in the same order as $\mathbf{g}_p(\boldsymbol{\eta})$. Let $\mathbf{u}_p(\boldsymbol{\eta}) = \mathbf{g}_p(\boldsymbol{\eta}) + \mathbf{b}_p$. Thus, the Jacobian of the parametric intensity mapping is

$$\mathbf{J}_p(\boldsymbol{\eta}) = 2 \text{Re} \left\{ \text{diag}(\mathbf{u}_p^*(\boldsymbol{\eta})) \frac{\partial \mathbf{g}_p(\boldsymbol{\eta})}{\partial \boldsymbol{\eta}^T} \right\} \in \mathbb{R}^{Q_p \times D_\eta}. \quad (45)$$

The same Jacobian governs local identifiability and GN conditioning. A small $\sigma_{\min}(\mathbf{J}_p)$ exposes weak parameter directions and unstable updates; phase-diverse references can make these directions observable.

3) *Damped Gauss–Newton Update*: At iteration i , define $\mathbf{u}_p^{(i)} = \mathbf{g}_p(\boldsymbol{\eta}^{(i)}) + \mathbf{b}_p$, $\mathbf{e}_p^{(i)} = \mathbf{z}_p - \left| \mathbf{u}_p^{(i)} \right|^2$, and $\mathbf{J}_p^{(i)} = \mathbf{J}_p(\boldsymbol{\eta}^{(i)})$ denotes the intensity-domain Jacobian evaluated at the current iterate.

The damped Gauss–Newton increment is then computed as

$$\Delta \boldsymbol{\eta}^{(i)} = \left[\left(\mathbf{J}_p^{(i)} \right)^T \mathbf{J}_p^{(i)} + \lambda^{(i)} \mathbf{I} \right]^{-1} \left(\mathbf{J}_p^{(i)} \right)^T \mathbf{e}_p^{(i)}, \quad (46)$$

Let $\sigma_{\max}^{(i)}$ and $\sigma_{\min}^{(i)}$ be the extreme singular values of $\mathbf{J}_p^{(i)}$. Theorem 2 motivates the following damping for a target condition number $\kappa_{\text{tar}} > 1$:

$$\begin{aligned} \lambda_{\text{stab}}^{(i)} &= \left[\frac{(\sigma_{\max}^{(i)})^2 - \kappa_{\text{tar}} (\sigma_{\min}^{(i)})^2}{\kappa_{\text{tar}} - 1} \right]_+, \\ \lambda^{(i)} &= \max\{\lambda_{\text{LM}}^{(i)}, \lambda_{\text{stab}}^{(i)}\}. \end{aligned} \quad (47)$$

Here $\lambda_{\text{LM}}^{(i)}$ follows the acceptance rule, while the stability floor caps the condition number of the regularized normal matrix by κ_{tar} ; it does not alter the rank or conditioning of the

underlying Jacobian. A persistently negligible $\sigma_{\min}^{(i)}$ instead signals insufficient pilot or reference diversity, which damping cannot restore. The update is

$$\boldsymbol{\eta}_{\text{trial}}^{(i+1)} = \Pi_{\Omega} \left(\boldsymbol{\eta}^{(i)} + \Delta \boldsymbol{\eta}^{(i)} \right), \quad (48)$$

where $\Pi_{\Omega}(\cdot)$ clips the angles to $\Theta \times \Phi$ and radially clips each complex gain at $\alpha_{\max}$. The trial point is accepted and λ_{LM} is divided by 3 when it lowers F_p ; otherwise, the current iterate is retained and λ_{LM} is multiplied by 10.

4) *Initialization*: Because (41) is nonconvex, we first obtain a coarse unstructured channel estimate from the calibrated spatial reference. Neglecting the second-order unknown-field term in the intensity expansion gives the linearized initializer

$$\mathbf{h}^{(0)} = \arg \min_{\mathbf{h}} \left\| \mathbf{z}_p - |\mathbf{b}_p|^2 - 2 \operatorname{Re} \{ \mathbf{b}_p^* \odot \mathbf{G}_p \mathbf{h} \} \right\|_2^2, \quad (49)$$

from which $\mathbf{H}^{(0)}$ is reconstructed. Although this approximation is most accurate when the reference term dominates locally, the subsequent damped nonlinear iterations use the complete intensity model.

The initial DOA of each user is then obtained through a coarse angular search,

$$\left(\theta_l^{(0)}, \phi_l^{(0)} \right) = \arg \max_{(\theta, \phi) \in \mathcal{G}_{\Omega}} \frac{|\mathbf{v}^H(\theta, \phi) \mathbf{h}_l^{(0)}|^2}{\|\mathbf{v}(\theta, \phi)\|_2^2}, \quad (50)$$

where $\mathcal{G}_{\Omega}$ denotes a coarse angular grid. Given the coarse DOA estimate, the corresponding complex path gain is obtained by least-squares projection onto the steering vector. These coarse parameter estimates are then used to initialize the damped Gauss–Newton iterations. The complete joint estimation procedure is summarized in Algorithm 1.

C. Finite-Alphabet Symbol Detection

After training, $\hat{\mathbf{H}}$ is used to construct $\hat{\mathbf{G}}_d$ in (13). Because $\mathbf{s}_d \in \mathcal{S}^{D_s}$ is finite-alphabet, reliability is governed by the energy-domain separation in (27), rather than by a continuous Jacobian rank condition.

1) *Maximum-Likelihood Detection*: Under the adopted independent Gaussian intensity-domain noise model with identical variance, the maximum-likelihood detector is equivalent to

$$\hat{\mathbf{s}}_{\text{ML}} = \arg \min_{\mathbf{s} \in \mathcal{S}^{D_s}} \left\| \mathbf{z}_d - |\hat{\mathbf{G}}_d \mathbf{s} + \mathbf{b}_d|^2 \right\|_2^2. \quad (51)$$

Replacing $\mathbf{G}_d$ in (27) by $\hat{\mathbf{G}}_d$ gives the estimated margin. Exhaustive ML still evaluates $|\mathcal{S}|^{D_s}$ candidates and is mainly a short-block benchmark.

2) *Linearized Initialization and Candidate Screening*: If the reference-induced interference dominates the unknown self-intensity, $\mathbf{z}_d - |\mathbf{b}_d|^2 \approx \mathbf{A}_d \bar{\mathbf{s}}_d$, where $\bar{\mathbf{s}} = [\operatorname{Re}\{\mathbf{s}\}^T, \operatorname{Im}\{\mathbf{s}\}^T]^T$ and

$$\mathbf{A}_d = 2 \left[\operatorname{Re}\{\operatorname{diag}(\mathbf{b}_d^*) \hat{\mathbf{G}}_d\} \quad -\operatorname{Im}\{\operatorname{diag}(\mathbf{b}_d^*) \hat{\mathbf{G}}_d\} \right]. \quad (52)$$

A regularized real-domain initialization is

$$\bar{\mathbf{s}}_{\text{lin}} = (\mathbf{A}_d^T \mathbf{A}_d + \gamma_{\text{lin}} \mathbf{I})^{-1} \mathbf{A}_d^T (\mathbf{z}_d - |\mathbf{b}_d|^2), \quad (53)$$

Algorithm 1 Joint DOA and Channel-Gain Estimation

Require: $\mathbf{z}_p$, $\mathbf{x}_p[t]$, $\mathbf{b}_p$, receiving-surface geometry, Ω , $\mathcal{G}_{\Omega}$, I_c , $\lambda_{\text{LM}}^{(0)}$, κ_{tar} , and $\epsilon_c, \epsilon_0 > 0$.

Ensure: Estimated parameter vector $\hat{\boldsymbol{\eta}}$ and channel $\hat{\mathbf{H}}$.

- 1: Obtain $\mathbf{H}^{(0)}$ from the linearized spatial-code initializer in (49).
 - 2: Obtain coarse DOAs using (50).
 - 3: Form $\boldsymbol{\eta}^{(0)}$ according to (39).
 - 4: **for** $i = 0, 1, \dots, I_c - 1$ **do**
 - 5: Compute $\mathbf{u}_p^{(i)} = \mathbf{g}_p(\boldsymbol{\eta}^{(i)}) + \mathbf{b}_p$.
 - 6: Compute the residual $\mathbf{e}_p^{(i)} = \mathbf{z}_p - \left| \mathbf{u}_p^{(i)} \right|^2$.
 - 7: Construct $\mathbf{J}_p^{(i)}$ using (45).
 - 8: Set $\lambda^{(i)}$ using the stability floor in (47).
 - 9: Compute $\Delta \boldsymbol{\eta}^{(i)}$ using (46).
 - 10: Compute the projected trial point using (48).
 - 11: **if** the trial point decreases F_p **then**
 - 12: Accept $\boldsymbol{\eta}^{(i+1)} = \boldsymbol{\eta}_{\text{trial}}^{(i+1)}$.
 - 13: Set $\lambda_{\text{LM}}^{(i+1)} = \lambda_{\text{LM}}^{(i)}/3$.
 - 14: **else**
 - 15: Set $\boldsymbol{\eta}^{(i+1)} = \boldsymbol{\eta}^{(i)}$.
 - 16: Set $\lambda_{\text{LM}}^{(i+1)} = 10\lambda_{\text{LM}}^{(i)}$.
 - 17: **end if**
 - 18: **if** $\|\Delta \boldsymbol{\eta}^{(i)}\|_2 / (\|\boldsymbol{\eta}^{(i)}\|_2 + \epsilon_0) < \epsilon_c$ **then**
 - 19: **break**
 - 20: **end if**
 - 21: **end for**
 - 22: Set $\hat{\boldsymbol{\eta}} = \boldsymbol{\eta}^{(i+1)}$.
 - 23: Reconstruct $\hat{\mathbf{H}} = \mathbf{H}(\hat{\boldsymbol{\eta}})$.
-

where $\gamma_{\text{lin}} \geq 0$. The complex estimate $[\mathbf{I}; j\mathbf{I}]\bar{\mathbf{s}}_{\text{lin}}$ is projected element-wise onto $\mathcal{S}$ and used either to initialize an iterative detector or to center a reduced ML candidate list.

Let $0 < \underline{d}_d^{(E)} \leq \hat{d}_{\min, d}^{(E)}$ be a certified bound obtained for the realized reference map from (27) or another valid restricted-distance bound. A candidate $\mathbf{s}_c$ is the unique ML solution whenever

$$r(\mathbf{s}_c) \triangleq \left\| \mathbf{z}_d - |\hat{\mathbf{G}}_d \mathbf{s}_c + \mathbf{b}_d|^2 \right\|_2 < \frac{\underline{d}_d^{(E)}}{2}. \quad (54)$$

Every other candidate then has residual at least $\underline{d}_d^{(E)} - r(\mathbf{s}_c) > r(\mathbf{s}_c)$ by the triangle inequality. The test is therefore applied to the linearized candidate and subsequently to the GS iterates.

3) *Reference-Assisted Gerchberg–Saxton Detection*: GS alternates between measurement-magnitude and signal-domain constraints. We adapt it to the affine model by including the reference in the forward field and removing it before back-projection.

Given the current symbol estimate $\mathbf{s}^{(i)}$, the predicted affine field is

$$\mathbf{u}_d^{(i)} = \hat{\mathbf{G}}_d \mathbf{s}^{(i)} + \mathbf{b}_d. \quad (55)$$

The measured-magnitude constraint is then enforced as

$$\tilde{\mathbf{u}}_d^{(i)} = \sqrt{[\mathbf{z}_d]_+} \odot \exp(j \angle \mathbf{u}_d^{(i)}), \quad (56)$$

where $[\mathbf{z}]_+ = \max(\mathbf{z}, \mathbf{0})$ is applied element-wise.

Algorithm 2 Reference-Assisted Gerchberg–Saxton Symbol Detection

Require: $\mathbf{z}_d, \hat{\mathbf{G}}_d, \mathbf{b}_d, \mathcal{S}, \underline{d}_d^{(E)}, \gamma, I_d$, and $\epsilon_d, \epsilon_0 > 0$.

Ensure: Detected symbol vector $\hat{\mathbf{s}}_d$.

- 1: Compute $\mathbf{r}_d = \sqrt{[\mathbf{z}_d]_+}$.
 - 2: Precompute $\mathbf{B}_d$ according to (57).
 - 3: Initialize $\mathbf{s}^{(0)}$ using a linearized estimate, the previous data block, or a random constellation vector.
 - 4: **for** $i = 0, 1, \dots, I_d - 1$ **do**
 - 5: Compute the predicted affine field using (55).
 - 6: Evaluate $r(\mathbf{s}^{(i)})$ in (54).
 - 7: **if** $r(\mathbf{s}^{(i)}) < \underline{d}_d^{(E)}/2$ **then**
 - 8: Output $\hat{\mathbf{s}}_d = \mathbf{s}^{(i)}$ and terminate.
 - 9: **end if**
 - 10: Apply the magnitude constraint using (56).
 - 11: Back-project using (58).
 - 12: Apply the constellation projection using (59).
 - 13: **if** $\|\mathbf{s}^{(i+1)} - \mathbf{s}^{(i)}\|_2 / (\|\mathbf{s}^{(i)}\|_2 + \epsilon_0) < \epsilon_d$ **then**
 - 14: **break**
 - 15: **end if**
 - 16: **end for**
 - 17: Output $\hat{\mathbf{s}}_d = \mathcal{Q}_\mathcal{S}(\mathbf{s}^{(i+1)})$.
-

The reference vector is then removed, and the resulting field is projected back onto the signal domain. To improve numerical stability when $\hat{\mathbf{G}}_d$ is ill-conditioned, define the regularized back-projection matrix

$$\mathbf{B}_d = (\hat{\mathbf{G}}_d^H \hat{\mathbf{G}}_d + \gamma \mathbf{I})^{-1} \hat{\mathbf{G}}_d^H, \quad (57)$$

where $\gamma \geq 0$ is a regularization parameter. When $\gamma = 0$ and $\hat{\mathbf{G}}_d$ has full column rank, $\mathbf{B}_d = \hat{\mathbf{G}}_d^\dagger$.

The continuous signal-domain update is

$$\tilde{\mathbf{s}}^{(i+1)} = \mathbf{B}_d(\tilde{\mathbf{u}}_d^{(i)} - \mathbf{b}_d). \quad (58)$$

The finite-alphabet structure is incorporated through

$$\mathbf{s}^{(i+1)} = \mathcal{Q}_\mathcal{S}(\tilde{\mathbf{s}}^{(i+1)}), \quad (59)$$

where $\mathcal{Q}_\mathcal{S}(\cdot)$ denotes element-wise nearest-neighbor quantization onto the constellation $\mathcal{S}$.

In the proposed detector, (59) is applied after every back-projection to exploit the finite-alphabet prior. The complete procedure is summarized in Algorithm 2.

$\mathbf{B}_d$ and the distance bound are precomputed per coherence block; well-separated blocks can therefore terminate after few GS iterations.

As an iterative baseline, projected Wirtinger flow (WF) [30] applies gradient steps to the same affine intensity loss, projects the continuous iterate onto the constellation convex hull, and uses final nearest-neighbor quantization. It uses the same linearized initialization and distance certificate as GS.

D. Computational Complexity and Implementation Considerations

Let I_c , I_d , and I_{WF} denote the numbers of channel-estimation, GS, and projected-WF iterations, respectively. The principal computational costs are summarized in Table I.

GN forms $\mathbf{J}_p^T \mathbf{J}_p$ and solves a $D_\eta \times D_\eta$ system, where $D_\eta = 4L \ll 2N_r L$. Forming $\mathbf{B}_d$ costs $\mathcal{O}(Q_d D_s^2 + D_s^3)$ once per coherence block. Given $\underline{d}_d^{(E)}$, each online certificate test has the same order as one forward projection. Exact construction by pairwise enumeration, however, costs $\mathcal{O}(Q_d |\mathcal{S}|^{2D_s})$ per channel and reference configuration; it is used here only for the short blocks in the simulations and is excluded from the online orders in Table I.

E. Cramér–Rao Lower Bound for DOA Estimation

The CRLB follows directly from the parametric training model in (40) and its Jacobian in (45). Under the effective intensity-domain approximation introduced in Section II, assume that the noise is white Gaussian, $\mathbf{w}_p \sim \mathcal{N}(\mathbf{0}, \sigma_p^2 \mathbf{I})$, with variance independent of the unknown parameters. Repartition the real parameter vector in (39) as $\boldsymbol{\eta} = [\boldsymbol{\psi}^T, \boldsymbol{\nu}^T]^T$, where $\boldsymbol{\psi} = [\boldsymbol{\theta}^T, \boldsymbol{\phi}^T]^T$ contains the DOAs and $\boldsymbol{\nu} = [\text{Re}\{\boldsymbol{\alpha}\}^T, \text{Im}\{\boldsymbol{\alpha}\}^T]^T$ contains the nuisance path gains. The Fisher information matrix (FIM) is then

$$\mathbf{F}(\boldsymbol{\eta}) = \frac{1}{\sigma_p^2} \mathbf{J}_p^T(\boldsymbol{\eta}) \mathbf{J}_p(\boldsymbol{\eta}), \quad (60)$$

with the column partition $\mathbf{J}_p = [\mathbf{J}_\psi \ \mathbf{J}_\nu]$. Eliminating the unknown gains gives the equivalent FIM for the DOAs,

$$\mathbf{F}_{\text{DOA}} = \mathbf{F}_{\psi\psi} - \mathbf{F}_{\psi\nu} \mathbf{F}_{\nu\nu}^\dagger \mathbf{F}_{\nu\psi} = \frac{1}{\sigma_p^2} \mathbf{J}_\psi^T \mathbf{P}_\nu^\perp \mathbf{J}_\psi, \quad (61)$$

where $\mathbf{P}_\nu^\perp = \mathbf{I} - \mathbf{J}_\nu \mathbf{J}_\nu^\dagger$, and $(\cdot)^\dagger$ denotes the Moore–Penrose inverse. Thus, only the component of an angle-induced intensity perturbation orthogonal to the gain perturbation subspace carries DOA information. If $\mathbf{F}_{\text{DOA}}$ is nonsingular, any unbiased DOA estimator satisfies

$$\text{Cov}(\hat{\boldsymbol{\psi}}) \succeq \mathbf{F}_{\text{DOA}}^{-1}. \quad (62)$$

The diagonal entries give the individual bounds for θ_l and ϕ_l . Thus, weak or rank-deficient directions of $\mathbf{P}_\nu^\perp \mathbf{J}_\psi$ respectively produce large or unbounded DOA limits, linking reference design directly to attainable angular accuracy.

V. SIMULATION RESULTS

Numerical results first validate the identifiability and accuracy limits in Section III, and then examine how the resulting stability and distance criteria improve the algorithms in Section IV. The remaining experiments examine angular separation and reference–signal energy balance.

Unless otherwise stated, the simulations follow the two-phase model in Section II with a two-user 8×8 RF holographic receiving surface operating at 3.5 GHz and half-wavelength element spacing. The complex path gains are $\alpha_1 = e^{j0.35}$ and $\alpha_2 = 0.85e^{-j0.55}$, normalized QPSK is used, and the reference amplitude is 1.5. Each fixed spatial reference map assigns phases from $\{0, \pi/2, \pi, 3\pi/2\}$ across the receiving units and is reused over time; one energy integration is taken per symbol ($M = 1$). Training and data observations follow (40) and (13) with independent real Gaussian effective output

TABLE I
COMPUTATIONAL COMPLEXITY OF THE MAIN RECOVERY ALGORITHMS

Method	Unknown dimension	Approximate complexity	Primary role
Joint damped GN	D_η	$\mathcal{O}(I_c(Q_p D_\eta^2 + D_\eta^3))$	DOA and channel-gain estimation
Exhaustive ML	D_s	$\mathcal{O}(S ^{D_s} Q_d D_s)$	Performance benchmark
Linearized detector	D_s	$\mathcal{O}(Q_d D_s^2 + D_s^3)$	Initialization and candidate screening
Projected WF	D_s	$\mathcal{O}(I_{WF} Q_d D_s)$	Iterative detection baseline
Reference-assisted GS	D_s	$\mathcal{O}(I_d Q_d D_s)$	Low-complexity symbol detection

noise of variance σ_w^2 per intensity sample. For a noiseless intensity vector $\boldsymbol{\mu} = |\mathbf{G}\boldsymbol{\vartheta} + \mathbf{b}|^2 \in \mathbb{R}^Q$, the SNR is defined as

$$\text{SNR} = 10 \log_{10} \left(\frac{\mathbb{E}\{\|\boldsymbol{\mu}\|_2^2\}}{Q\sigma_w^2} \right). \quad (63)$$

Here, the expectation is evaluated over the noiseless samples of the corresponding experiment and, in the data phase, over equiprobable transmitted symbol vectors. Scenario-specific DOAs, pilot lengths, and reference or surface configurations are stated with the corresponding experiments.

A. Channel Estimation Performance

For N_{MC} Monte Carlo trials, the reported channel normalized mean-square error is

$$\text{NMSE}_{\mathbf{H}} = \frac{1}{N_{MC}} \sum_{r=1}^{N_{MC}} \frac{\|\hat{\mathbf{H}}^{(r)} - \mathbf{H}\|_F^2}{\|\mathbf{H}\|_F^2}. \quad (64)$$

Its value in decibels is $\text{NMSE}_{\mathbf{H},\text{dB}} = 10 \log_{10} \text{NMSE}_{\mathbf{H}}$. Let $\boldsymbol{\psi}_l = [\theta_l, \varphi_l]^T$ and $\hat{\boldsymbol{\psi}}_l^{(r)}$ denote the true and estimated angular vectors of user l in degrees. The reported DOA RMSE averages all $2L$ angular components:

$$\text{RMSE}_{\text{DOA}} = \left[\frac{1}{2N_{MC}L} \sum_{r=1}^{N_{MC}} \sum_{l=1}^L \left\| \hat{\boldsymbol{\psi}}_l^{(r)} - \boldsymbol{\psi}_l \right\|_2^2 \right]^{1/2}. \quad (65)$$

The root CRLB uses the same aggregation. With $T_p = 32$ QPSK pilots and user DOAs $(15^\circ, 8^\circ)$ and $(35^\circ, -10^\circ)$, Fig. 3 shows that the RMSE decreases from 1.04° at 0 dB to 0.060° at 24 dB and closely follows the CRLB over 1000 noise realizations. Because the constrained estimator may be biased at finite SNR, the CRLB is used as a local accuracy benchmark.

Figure 4 next examines the counting hierarchy in (26) for the per-element $L = 2$ unstructured training problem using nested QPSK prefixes and 10,000 paired trials. With a fixed reference, the noiseless recovery probability rises from about 0.54 at $T_p = 2L$ to above 0.99 at $T_p = 3L$, whereas the no-reference model retains its global-phase ambiguity and an approximately 3 dB NMSE floor. Beyond the threshold, additional pilots improve robustness, reaching about -19.3 dB NMSE at $T_p = 48$; isolated success below $3L$ does not imply global injectivity.

Beyond the counting threshold, the required training length depends on local stability. Figure 5 compares the proposed Jacobian-CRLB policy with fixed-length, residual-based, and oracle stopping. All policies use nested QPSK prefixes, the practical initialization in Algorithm 1, and LM refinement. For users at polar angles 19° and 21° with a common 5°

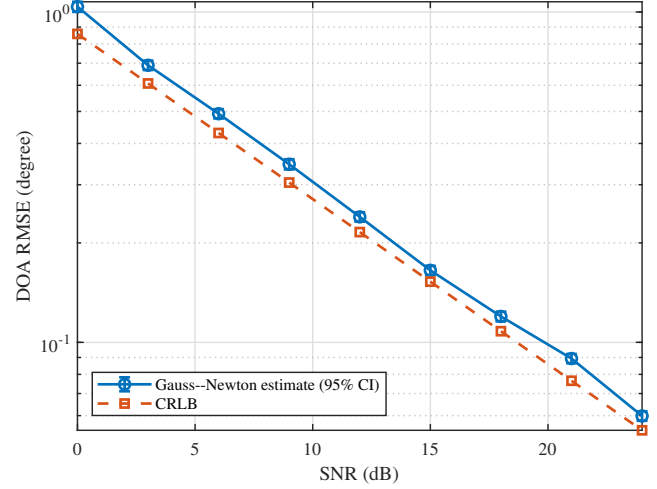


Fig. 3. Per-angle DOA RMSE and root CRLB versus SNR. Error bars show 95% confidence intervals.

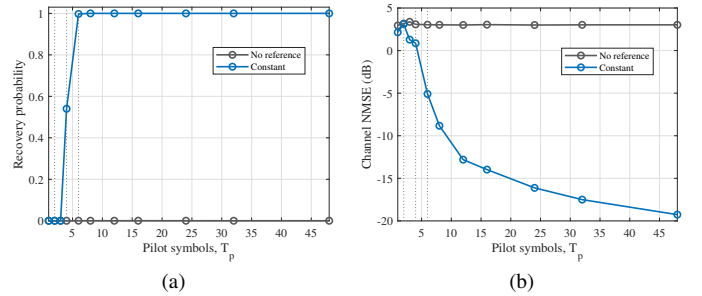


Fig. 4. Pilot-count dependence of (a) noiseless multistart recovery probability and (b) channel NMSE at 12 dB. The vertical lines mark $T_p = L, 2L$, and $3L$.

azimuth, the candidate lengths are $T_p \in \{8, 12, \dots, 40\}$, with 300 paired trials per SNR. The proposed policy selects the shortest prefix satisfying a -25 dB channel CRLB-NMSE target, a 1° DOA-CRLB target, and a 5×10^3 condition-number limit.

The residual rule stops near $T_p = 8$ even when the physical parameters remain inaccurate: at 8 dB it uses 8.33 pilot symbols and yields -22.22 dB NMSE. The proposed policy instead selects 30.92 symbols and reaches -28.16 dB. At 10 dB it uses 20.04 symbols to attain -28.13 dB, and at high SNR it approaches the shortest prefix. Fixed-long training gives the lowest NMSE but always uses 40 symbols. Thus, the stability test identifies information insufficiency that a small residual alone does not reveal.

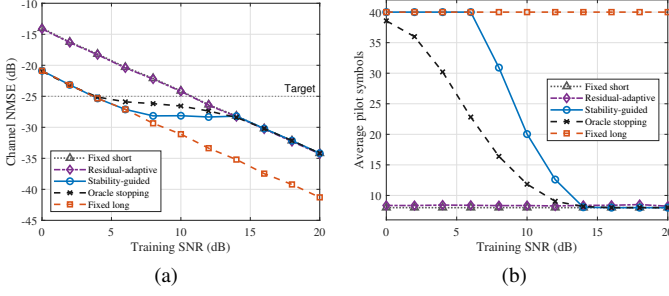


Fig. 5. Stability-guided adaptive training versus SNR: (a) channel NMSE and (b) average selected pilot length.

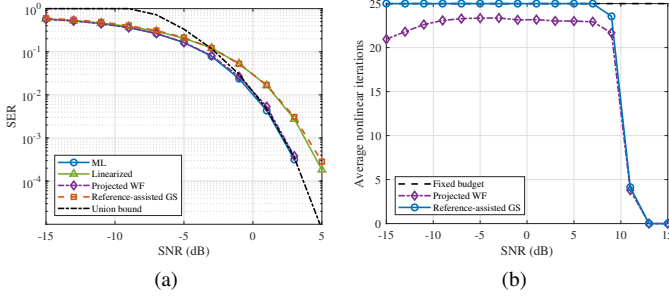


Fig. 6. Perfect-CSI detection: (a) detector SER and analytical union bound and (b) average nonlinear iterations under the $d_{\min}^{(E)}$ certificate.

B. Symbol Detection Performance with Perfect CSI

For N_{MC} trials, the symbol error rate used throughout the following experiments is

$$\text{SER} = \frac{1}{N_{MC} L K_d} \sum_{r=1}^{N_{MC}} \sum_{l=1}^L \sum_{k=1}^{K_d} \mathbf{1}_{\{\hat{s}_{l,k}^{(r)} \neq s_{l,k}^{(r)}\}}.$$

We isolate the detector from channel-estimation errors by constructing $\mathbf{G}_d$ from the true channel. Two QPSK users and the simultaneous spatial four-phase map with $\rho = 1.5$ are used. Figure 6(a) compares the detectors over 30,000 symbol-vector trials per point. At -5 dB, the ML, projected-WF, and certified-GS SERs are 0.164, 0.165, and 0.214, respectively. The distance-spectrum union bound derived from Theorem 3 approaches ML in the waterfall region.

Figure 6(b) evaluates the stopping test in (54) using the same trials. At low SNR the certificate is deliberately inactive and both iterative methods approach the 25-iteration budget; at 11 dB, their averages fall to 3.80 for projected WF and 4.13 for GS. Projected WF additionally requires a gradient evaluation and backtracking at each iteration.

To verify the pairwise theory directly, Fig. 7(a) uses 32 equal-power fixed spatial phase maps spanning a broad range of normalized minimum distances. The closest-pair Monte Carlo results follow (29), validating $d_{\min}^{(E)}$ as a reference-design metric.

Figure 7(b) further tests modulation-order robustness under the same fixed spatial four-phase map and perfect CSI. The QPSK and square-QAM constellations have unit average energy, and each point contains 10,000 trials. The waterfall shifts right as the constellation becomes denser; at 12 dB, the 16-

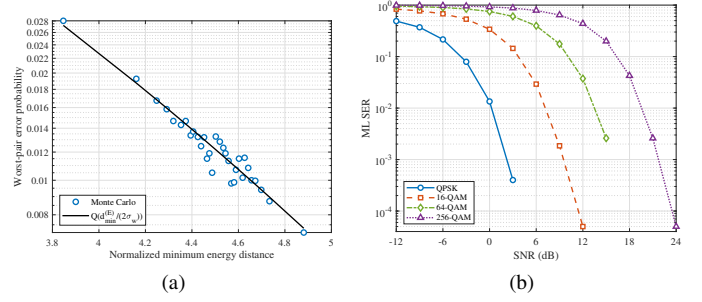


Fig. 7. Finite-alphabet reliability under perfect CSI: (a) worst-pair error probability versus normalized minimum energy distance and (b) exact-ML SER for different modulation alphabets.

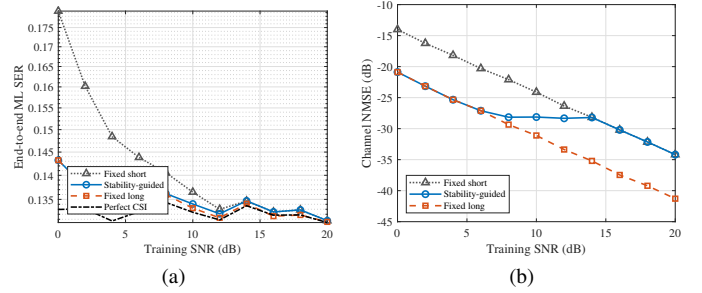


Fig. 8. End-to-end performance versus training SNR: (a) SER and (b) channel NMSE for fixed-length and stability-guided training.

, 64-, and 256-QAM SERs are 5×10^{-5} , 3.74×10^{-2} , and 0.436, respectively.

C. End-to-End Symbol Detection with Estimated CSI

We next propagate the preceding channel estimates to exhaustive data detection at 0 dB data SNR. Each of 300 training-noise realizations uses 200 data vectors. The fixed-short, stability-guided, and fixed-long receivers differ only in their pilot prefixes, while the perfect-CSI curve gives the data-noise floor.

At low training SNR, the stability rule requests the long prefix and closely tracks fixed-long training. At 0 dB, it lowers the SER from about 0.18 with eight pilots to about 0.14 and improves the channel NMSE from -14.0 to -20.9 dB. At higher training SNR, the selected prefix shortens and the estimated-CSI curves approach the perfect-CSI floor, confirming that the gain comes from controlling channel-error propagation.

D. Effect of Angular Separation

Figure 9 connects spatial conditioning to detection. Increasing the separation from 0.25° to 8° raises the normalized minimum singular value from 0.0209 to 0.609 and lowers the perfect-CSI SER from 0.409 to 0.248 at -6 dB. The mild nonmonotonicity at larger separations reflects finite-aperture sidelobes.

The result confirms that the array manifold must preserve the relevant field differences before the reference-assisted projection can separate their intensities.

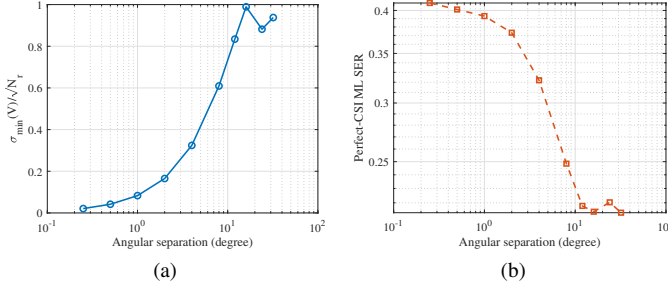


Fig. 9. User angular separation versus (a) array-manifold conditioning and (b) perfect-CSI ML SER at -6 dB.

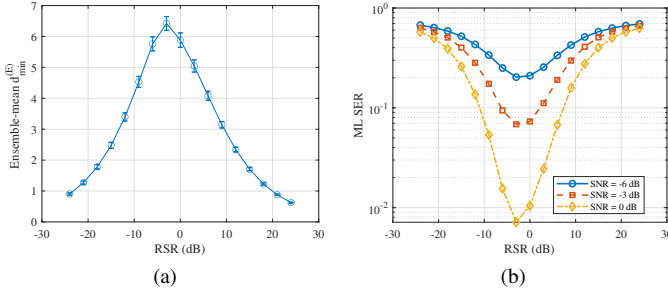


Fig. 10. Reference-to-signal energy ratio under a fixed total-energy budget: (a) ensemble-mean minimum energy distance and (b) perfect-CSI ML SER. Error bars show one standard deviation across phase maps.

E. Effect of the Reference-to-Signal Energy Ratio

We finally examine the energy allocation in Corollary 2. The total average incident energy per unit is fixed as $P_s + P_b = 1$, so varying $\text{RSR} = 10 \log_{10}(P_b/P_s)$ changes only the allocation. Figure 10 averages 1500 trials over each of 24 phase-balanced fixed maps; the noise variance is calibrated at $\text{RSR} = 0$ and held fixed over the sweep.

Figure 10(a) shows that the ensemble-mean minimum distance is maximized near balance, at the sampled value $\text{RSR} = -3$ dB for this scenario. Its displacement from 0 dB agrees with (36), whose optimum depends on A_{ij}/B_{ij} , and the SER curves exhibit the corresponding bowl shape. Thus, severe imbalance weakens phase-sensitive separation, while the preferred ratio remains scenario dependent.

VI. CONCLUSION

This paper developed a reference-assisted RF holographic receiver for recovering complex channel and data information from intensity-only measurements. For the resulting affine intensity model, we established Jacobian-based uniqueness and stability conditions for continuous channel sensing and an energy-domain minimum-distance criterion for finite-alphabet detection. The analysis further revealed the roles of pilot diversity, spatial four-phase reference coding, and reference-signal energy balance. Guided by these results, we designed a two-stage receiver combining linearized initialization, stability-guided adaptive training, damped Gauss-Newton estimation, and distance-certified GS detection. Simulations under representative two-user far-field settings supported the predicted trends and showed that the adaptive scheme reduced the

average pilot length from 40 to 20.0 at 10 dB while attaining -28.1 dB channel NMSE. These results demonstrate the feasibility of recovering complex communication information without coherent I/Q acquisition.

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